Graph Theory & Networks (AI HL)

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Variables x and y satisfy $y = ax^b$. A graph of $\ln y$ against $\ln x$ is a straight line passing through $(0, 1.20)$ and $(4, 6.80)$. Find the values of the constants a and b to 3 significant figures.

Take natural logarithms of $y = ax^b$: $\ln y = \ln a + b\ln x$ This is a linear equation $Y = mX + c$ with $Y = \ln y$, $X = \ln x$, gradient $m = b$, and intercept $c = \ln a$. Gradient: $b = \frac{6.80 - 1.20}{4 - 0} = \frac{5.60}{4} = 1.40$ Vertical intercept gives $\ln a = 1.20$: $a = e^{1.20} \approx 3.32$ Therefore, $a = 3.32$ and $b = 1.40$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Given $\ln y = 2\ln x + \ln 3$, express y in terms of x. Hence sketch the graph.

$\ln y = \ln x^2 + \ln 3 = \ln(3x^2) \Rightarrow y = 3x^2$. Parabola through origin, scaled by 3.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A student calculates $\int_1^3 \frac{4}{2x+1}\,dx$ by rounding $\ln 7 \approx 1.95$ and $\ln 3 \approx 1.10$ at the start, obtaining $2(1.95 - 1.10) = 1.70$. State the exact value of the integral and calculate the absolute error introduced by premature rounding.

Integrate exactly: $\int_1^3 \frac{4}{2x+1}\,dx = [2\ln(2x+1)]_1^3 = 2\ln 7 - 2\ln 3 = 2\ln\left(\frac{7}{3}\right) \approx 1.6946\ldots$ The exact value is $2\ln\left(\frac{7}{3}\right)$. The premature rounding gives 1.70, yielding an absolute error of: $|1.70 - 1.6946| \approx 0.0054$ On Cambridge Paper 3, this premature rounding costs the final accuracy mark (A1) because the 3 s.f. answer is 1.69, not 1.70.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 2 · Medium Use the method of cylindrical shells to find the volume generated by revolving the region bounded by $y = 3x - x^2$ and $y = 0$ about the y-axis.

Find intersections with $y = 0$: $3x - x^2 = 0 \Rightarrow x(3 - x) = 0$, so $a = 0$ and $b = 3$. Revolving about y-axis: shell radius $r(x) = x$, shell height $h(x) = 3x - x^2$. $V = 2\pi\int_0^3 x(3x - x^2)\,dx = 2\pi\int_0^3 (3x^2 - x^3)\,dx = 2\pi\left[x^3 - \frac{x^4}{4}\right]_0^3$ $= 2\pi\left(27 - \frac{81}{4}\right) = 2\pi\left(\frac{27}{4}\right) = \frac{27\pi}{2}$

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5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.

Complete the square on the y-terms: $y^2 - 4y = -8x - 28 \Rightarrow (y^2 - 4y + 4) = -8x - 28 + 4$.
$(y-2)^2 = -8x - 24 = -8(x+3)$. This matches $(y-k)^2 = 4p(x-h)$ with:Vertex: $(h,k) = (-3, 2)$
Focal parameter: $4p = -8 \Rightarrow p = -2$ (opens horizontally to the left)Focus:
$(h+p, k) = (-3 + (-2), 2) = (-5, 2)$Directrix: $x = h - p = -3 - (-2) \Rightarrow x = -1$

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force F N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.

(a) $F_f = 0.4 \times 3 \times 9.8 = 11.76$ N. (b) $a = \frac{20 - 11.76}{3} \approx 2.75$ m s^{-2}. (c) $t = \sqrt{10/2.75} \approx 1.91$ s. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ for the function: $y = \int_{\cos x}^{x^3} \sqrt{1 - t^2}\, dt$.

Apply the Leibniz Rule with upper limit $u(x) = x^3 \Rightarrow u'(x) = 3x^2$ and lower limit
$v(x) = \cos x \Rightarrow v'(x) = -\sin x$: $\frac{dy}{dx} = \sqrt{1 - (x^3)^2} \cdot (3x^2) - \sqrt{1 - (\cos x)^2} \cdot (-\sin x)$. Simplify trigonometric terms with $\sqrt{1 - \cos^2 x} = \sqrt{\sin^2 x} = |\sin x|$: $\frac{dy}{dx} = \mathbf{3x^2\sqrt{1 - x^6} + |\sin x|\sin x}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · Foundations Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$.

Find intersection points: $x^2 = 2x - x^2 \Rightarrow 2x^2 - 2x = 0 \Rightarrow 2x(x-1) = 0$, so $x = 0$ and $x = 1$. On $[0, 1]$, test $x = 0.5$: $2(0.5) - (0.5)^2 = 0.75 > (0.5)^2 = 0.25$, so $y = 2x - x^2$ is the upper curve.
$A = \int_0^1 [(2x - x^2) - x^2]\, dx = \int_0^1 (2x - 2x^2)\, dx = \left[x^2 - \frac{2x^3}{3}\right]_0^1 = 1 - \frac{2}{3} = \frac{1}{3}$

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the set of values of x satisfying the inequality $2x^2 - 7x + 3 \leq 0$.

Find the critical values where $2x^2 - 7x + 3 = 0$. Factor the quadratic: $(2x-1)(x-3) = 0$, so the roots are $x = \frac{1}{2}$ and $x = 3$. Since the coefficient of x^2 is positive ($a = 2 > 0$), the parabola opens upward. The quadratic is less than or equal to zero between and including the roots. Therefore, the solution set is $\frac{1}{2} \leq x \leq 3$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The equation $x^3 - x - 1 = 0$ has a root near $x = 1.3$. (a) Show that the iterative formula $x_{n+1} = (x_n + 1)^{1/3}$ converges by considering $g'(x)$. (b) Starting with $x_1 = 1.3$, calculate x_2 and x_3 to 4 decimal places.

(a) Let $g(x) = (x + 1)^{1/3}$. Differentiate: $g'(x) = \frac{1}{3}(x + 1)^{-2/3} = \frac{1}{3(x+1)^{2/3}}$ Near the root $x \approx 1.3$: $g'(1.3) = \frac{1}{3(2.3)^{2/3}} \approx \frac{1}{3(1.742)} \approx 0.191$ Since \(|g'(x)| $x_1 = 1.3$ $x_2 = (1.3 + 1)^{1/3} = (2.3)^{1/3} \approx 1.3200$ $x_3 = (1.3200 + 1)^{1/3} = (2.3200)^{1/3} \approx 1.3238$

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 1 · Foundations Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = abla(x^3y + 2z)$ along any curve C from $(0, 0, 1)$ to $(2, 1, 3)$.

The field is conservative with potential function $f(x, y, z) = x^3y + 2z$. By the Fundamental Theorem for Line Integrals: $\int_C ablaf \cdot d\mathbf{r} = f(2, 1, 3) - f(0, 0, 1)$ Evaluate at endpoints: $f(2, 1, 3) = 2^3(1) + 2(3) = 8 + 6 = 14$ $f(0, 0, 1) = 0^3(0) + 2(1) = 2$ Therefore: $\int_C \mathbf{F} \cdot d\mathbf{r} = 14 - 2 = 12$

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 1 · FoundationsFind the exact natural domain of the function $g(x) = \sqrt{\frac{x-1}{x+3}} - \ln(5 - x)$ in interval notation.

We must evaluate two simultaneous restrictions:Square root radicand must be non-negative and denominator non-zero: $\frac{x-1}{x+3} \geq 0 \quad \text{and} \quad x + 3eq0.$ Sign analysis of the quotient with critical numbers $x = -3$ and $x = 1$: • For \(x Logarithm argument must be strictly positive: x

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$.

Step 1: Identify progression parameters: First term $a_1 = 4$, common difference $d = 9 - 4 = 5$. Step 2: Compute a_{15}: $a_{15} = a_1 + (15 - 1)d = 4 + 14(5) = 4 + 70 = 74$. Step 3: Compute finite sum S_{20}:

$S_{20} = \frac{20}{2}[2a_1 + (20 - 1)d] = 10[2(4) + 19(5)] = 10[8 + 95] = 10(103) = 1030.$

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 1 · FoundationsFind $h'(2)$ given the accumulation function $h(x) = \int_0^{x^2} \frac{t}{t+1}\, dt$.

By FTC Part 1 and the Chain Rule, let $u = x^2 \Rightarrow \frac{du}{dx} = 2x$: $h'(x) = \frac{u}{u+1} \cdot \frac{du}{dx} = \frac{x^2}{x^2+1} \cdot (2x) = \frac{2x^3}{x^2+1}$. Evaluate at $x = 2$: $h'(2) = \frac{2(2^3)}{2^2+1} = \frac{2(8)}{4+1} = \mathbf{\frac{16}{5}}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

A student computes the volume of a cylinder using $r = 3.2$ cm and $h = 7.85$ cm. The raw calculator display shows $252.518...$. How should the final answer be reported according to Cambridge AS Level guidelines?

Unless otherwise specified, non-exact numerical answers must be given to 3 significant figures. Rounding $252.518...$ to 3 s.f. gives $253\ \text{cm}^3$. Giving 1 or 2 s.f., or excessive decimal places without justification, incurs an accuracy penalty (A1).

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains 250 mg, how much remains after 40 years?

(a) Half-life calculation: $t_{1/2} = \frac{\ln 2}{k} = \frac{0.693147}{0.035} \approx 19.8$ years. (b) Remaining mass at $t = 40$:

$A(t) = A_0 e^{-kt} = 250e^{-0.035(40)} = 250e^{-1.4} \approx 250(0.246597) \approx 61.65$ mg.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Matrices (AI HL)

$f(x) = cx^2$ for $0 \le x \le 3$, zero otherwise. Find c, $E(X)$, and $P(1 \le X \le 2)$.

$\int_0^3 cx^2\,dx = 9c = 1 \Rightarrow c = \frac{1}{9}$. $E(X) = \frac{1}{9}\int_0^3 x^3\,dx = \frac{1}{9}\cdot\frac{81}{4} = \frac{9}{4}$. $P(1 \le X \le 2) = \frac{1}{9}\left[\frac{x^3}{3}\right]_1^2 = \frac{1}{27}(8-1) = \frac{7}{27}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.03 Markov Chains (AI HL)

A manufacturer claims that the mean breaking strength of cables is at least 80 kN. A sample of 16 cables gives $\bar{x} = 77.5$ kN with population SD 5 kN. Test this claim at 1% significance.

$H_0 : \mu = 80$, $H_1 : \mu < 80$. $Z = \frac{77.5 - 80}{5/4} = -\frac{2.5}{1.25} = -2$. Critical value at 1%: -2.326. Since $-2 > -2.326$, do not reject H_0. Insufficient evidence at 1% that mean breaking strength is below 80 kN.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Proof by Mathematical Induction

A discrete random variable Y has $E(Y) = 5$ and $\text{Var}(Y) = 4$. Calculate $E(Y^2)$ and $\text{Var}(10 - 3Y)$.

Using $\text{Var}(Y) = E(Y^2) - [E(Y)]^2$, we get $E(Y^2) = \text{Var}(Y) + [E(Y)]^2 = 4 + 5^2 = 4 + 25 = 29$. For the linear transformation: $\text{Var}(10 - 3Y) = (-3)^2\text{Var}(Y) = 9 \times 4 = 36$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Level 1 · Foundations Find $\frac{dy}{dx}$ in terms of t for the parametric curve $x = 2\cos t$, $y = 5\sin t$ for $t \in (0, \pi)$.

Compute derivatives with respect to t: $\frac{dx}{dt} = -2\sin t$ $\frac{dy}{dt} = 5\cos t$ By Theorem 9.1: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{5\cos t}{-2\sin t} = -\frac{5}{2}\cot t$

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What does the Handshaking Lemma state for any undirected graph with V vertices and E edges?

a. The sum of the degrees of all vertices equals twice the number of edges: $\sum \deg(v) = 2E$ ✓

b. The sum of the degrees equals E

c. The number of vertices with odd degree is always odd

d. Every vertex must have degree at least 2

Because every edge connects two endpoints, each edge contributes exactly 2 to the sum of the degrees of the vertices: $\sum \deg(v) = 2E$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A graph has 6 vertices, each having degree 3. How many edges does the graph have?

a. 9 ✓

b. 18

c. 6

d. 12

$\sum \deg(v) = 6 \times 3 = 18$. By the Handshaking Lemma, $2E = 18 \Rightarrow E = 9$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Why can a graph never have an odd number of vertices of odd degree?

a. Because $\sum \deg(v) = 2E$ is always even, the sum of odd degrees must be even, forcing an even count of odd vertices ✓

b. Because edges cannot cross each other

c. By Euler's formula $V - E + F = 2$

d. Because every vertex must have at least one adjacent edge

The total sum of degrees is even ($2E$). The sum of even degrees is even, so the sum of the odd degrees must also be even, which requires an even number of odd terms.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What condition is both necessary and sufficient for a connected graph to contain an Eulerian circuit (a closed trail visiting every edge exactly once)?

a. Every vertex has an even degree ✓

b. Exactly two vertices have odd degree

c. The graph contains no cycles

d. The graph is a complete bipartite graph

Euler proved that a connected graph has an Eulerian circuit if and only if all vertices have even degree.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If a connected graph has exactly two vertices of odd degree, what type of Eulerian walk does it contain?

a. An Eulerian trail (open path visiting every edge once) that starts at one odd vertex and ends at the other ✓

b. An Eulerian circuit returning to the start

c. No Eulerian trail or circuit can exist

d. A Hamiltonian cycle

Euler's theorem guarantees an open Eulerian trail if and only if exactly two vertices have odd degree; the trail must start at one odd vertex and terminate at the other.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the difference between an Eulerian circuit and a Hamiltonian cycle?

a. An Eulerian circuit visits every edge once; a Hamiltonian cycle visits every vertex once ✓

b. An Eulerian circuit visits every vertex once; a Hamiltonian cycle visits every edge once

c. They are identical concepts in graph theory

d. Hamiltonian cycles apply only to directed trees

Eulerian paths/circuits pertain to visiting every edge, while Hamiltonian paths/cycles pertain to visiting every vertex.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a connected weighted graph with V vertices, how many edges does any spanning tree contain?

a. $V - 1$ ✓

b. V

c. $V + 1$

d. $\frac{V(V-1)}{2}$

Any tree on V vertices is minimally connected and contains no cycles, having precisely $V - 1$ edges.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How does Kruskal's algorithm select edges to construct a Minimum Spanning Tree (MST)?

a. **Sorts all edges by weight and greedily adds the smallest available edge that does not create a cycle** ✓

b. Grows a single tree by continuously adding the closest vertex to the current tree

c. Removes the largest cycle edges one by one until only a tree remains

d. Visits vertices in alphabetical order

Kruskal's algorithm is an edge-based greedy algorithm that adds edges in order of increasing weight, rejecting any edge that completes a cycle.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How does Prim's algorithm differ conceptually from Kruskal's algorithm?

a. **Prim grows a single connected tree from an arbitrary starting vertex, whereas Kruskal maintains a forest of components until unified** ✓

b. Prim works only on unweighted graphs

c. Kruskal finds the longest path rather than minimum tree

d. Prim can only be applied to planar graphs

Prim's algorithm builds a single growing connected tree vertex by vertex, whereas Kruskal's algorithm connects disjoint components across the whole graph.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the purpose of Dijkstra's algorithm?

a. To find the shortest paths from a single source vertex to all other vertices in a network with non-negative edge weights ✓

b. To find a minimum spanning tree

c. To determine if a graph has an Eulerian circuit

d. To find the maximum flow through a network

Dijkstra's algorithm is the standard greedy algorithm for computing single-source shortest paths on graphs with non-negative edge weights.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Why does Dijkstra's algorithm fail if a network contains negative edge weights?

a. A greedy choice once finalized could be improved later by a negative edge, invalidating the optimality assumption ✓

b. Distances cannot be negative by definition

c. The algorithm enters an infinite cycle immediately

d. Square roots of negative weights cannot be computed

Dijkstra's greedy assumption assumes that once a vertex distance is finalized, no shorter path to it exists; negative edges violate this monotonic property.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a planar graph, if $V = 6$ and $E = 9$, how many faces F does the graph enclose according to Euler's formula $V - E + F = 2$?

a. 5 ✓

b. 3

c. 4

d. 6

By Euler's planar formula $V - E + F = 2$: $6 - 9 + F = 2 \Rightarrow -3 + F = 2 \Rightarrow F = 5$.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $f''(x) > 0$ on an interval, the graph of f is:

a. **concave up ✓**

b. decreasing

c. concave down

d. linear

A positive second derivative means the graph curves upward (concave up).

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $f(x) = (x - 2)^2(x + 1)$, does the graph touch or cross the x-axis at $x = 2$?

a. The graph does not intersect the x-axis there

b. Not enough information

c. **Touches (does not cross) ✓**

d. Crosses

Multiplicity of $x = 2$ is 2 (even), so the graph touches the x-axis.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?

a. **4 ✓**

b. 2

c. 3

d. 6

The slope of the tangent line to the graph of $f(x)$ at any point is given by its derivative, $f'(x)$. 1. Find the derivative of $f(x) = x^2 - 2x - 1$ using the power rule: $f'(x) = 2x - 2$ 2. Evaluate the derivative at the x -coordinate of point $P(3, 2)$, which is $x = 3$: $f'(3) = 2(3) - 2 = 6 - 2 = 4$ Thus, the slope of the tangent line at P is 4. Misconception Analysis: - 2 results from mistakenly reading the y-coordinate of point P as the slope, since $f(3) = 2$. - 3 results from taking the x-c.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown. The vertex of the parabola is at the point $(3, -5)$. Find the value of the parameter k.

a. -5

b. 4 ✓

c. 9

d. 0

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. For the given function $f(x) = x^2 - 6x + k$, we have $a = 1$ and $b = -6$. Thus, the x-coordinate of the vertex is $x = -\frac{-6}{2(1)} = 3$. This matches the given vertex $(3, -5)$. Since the vertex $(3, -5)$ lies on the graph of the function, it must satisfy the function's equation. Substitute $x = 3$ and $f(x) = -5$ into the equation: $-5 = (3)^2 - 6(3) + k$ $-5 = 9 - 18 + k$ $-5 = -9 + k$ To solve for k, add 9 to both sides.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.

a. **1** ✓

b. 2

c. 3

d. 4

The equation of the tangent line to $f(x)$ at $x = 1$ is given as $y = 2x + c$. We are given that $f(1) = 3$. This means the point $(1, 3)$ lies on both the function $f(x)$ and its tangent line at $x = 1$. Since the point $(1, 3)$ lies on the tangent line $y = 2x + c$, we can substitute the coordinates into the equation: $3 = 2(1) + c$ $3 = 2 + c$ $c = 3 - 2$ $c = 1$ The derivative of the function at $x = 1$ is the slope of the tangent line, which is 2. This information is consistent with the problem but not direct.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.03 Markov Chains (AI HL)

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?

a. $(-2, 3)$

b. $(0, -1)$

c. $(1, 0)$ ✓

d. $(3, 4)$

The graph of $f(x)$ passes through the point $(1, 0)$. To verify this, we can observe the graph and identify the coordinates where the curve intersects the x-axis at $x = 1$. Let's check the other options. The point $(-2, 3)$ is not on the curve as the function value at $x = -2$ is approximately 5. The point $(0, -1)$ is not on the curve as the function value at $x = 0$ is approximately 3. The point $(3, 4)$ is not on the curve as the function value at $x = 3$ is approximately 0. Therefore, only $(1, 0)$ lies on the.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Proof by Mathematical Induction

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The function has a local minimum. What are the coordinates of the local minimum?

a. (2, 1)

b. **(2, -1)** ✓

c. (1, 0)

d. (3, 0)

The function is a parabola opening upwards, so its local minimum is also its vertex. The x-coordinate of the vertex of a parabola given by $f(x) = ax^2 + bx + c$ is $x = -b/(2a)$. For $f(x) = x^2 - 4x + 3$, we have $a = 1$ and $b = -4$. Therefore, the x-coordinate of the vertex is $x = -(-4)/(2 * 1) = 4/2 = 2$. To find the y-coordinate, we substitute $x = 2$ into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Thus, the coordinates of the local minimum are (2, -1).

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Consider the graph of the function $f(x)$ shown below. What is the limit of $f(x)$ as x approaches 1?

a. **1** ✓

b. 0

c. 2

d. Does not exist

The limit of a function as x approaches a certain value is the value that $f(x)$ gets arbitrarily close to as x gets arbitrarily close to that value. Observing the graph, as x approaches 1 from both the left and the right, the value of $f(x)$ approaches 1. Therefore, $\lim_{x \to 1} f(x) = 1$.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Consider the function $f(x)$ whose graph is given below. The function is defined piecewise. For $x \leq 2$, $f(x) = ax + b$. For $x > 2$, $f(x) = cx + d$. The graph of $f(x)$ passes through the point $(0, 3)$. The limit of $f(x)$ as x approaches 2 exists. Find the value of a.

a. -1 ✓

b. 1

c. 2

d. -2

For the limit of $f(x)$ as x approaches 2 to exist, the left-hand limit must equal the right-hand limit. That is, $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x)$. For $x \leq 2$, $f(x) = ax + b$. So, the left-hand limit is $\lim_{x \to 2^-} (ax + b) = 2a + b$. For $x > 2$, $f(x) = cx + d$. So, the right-hand limit is $\lim_{x \to 2^+} (cx + d) = 2c + d$. Therefore, $2a + b = 2c + d$. From the graph, we can see that the function passes through $(0, 3)$. This means $f(0) = 3$. Since $0 \leq 2$, we u.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.04 Partial Fractions

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

For a rational function $f(x) = \frac{P(x)}{Q(x)}$, vertical asymptotes occur at the zeros of the denominator $Q(x)$ that are not also zeros of the numerator $P(x)$. Given a vertical asymptote at $x = 1$, we know that $x = 1$ is a root of the denominator $x^2 - bx + 2 = 0$. Therefore, $1^2 - b(1) + 2 = 0$, which simplifies to $1 - b + 2 = 0$, so $b = 3$. The denominator is $x^2 - 3x + 2 = (x - 1)(x - 2)$. The horizontal asymptote is determined by the ratio of the leading coefficients of the numerator and denominator if t.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. a = 5, b = 3 ✓

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

For a rational function to have a removable discontinuity at $x = c$, the numerator must have a factor of $(x - c)$ that cancels with a factor in the denominator. Given a removable discontinuity at $x = 2$, the numerator $x^2 - ax + 6$ must have $x = 2$ as a root. Thus, $2^2 - a(2) + 6 = 0$, which simplifies to $4 - 2a + 6 = 0$, so $10 - 2a = 0$, and $a = 5$. The numerator can be factored as $(x - 2)(x - 3)$ since $2 \times 3 = 6$ and $2 + 3 = 5$, so $a = 5$. For the function to have a vertical asymptote at $x = 3$, the.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.

a. $a = 1, b = -4, c = 3$ ✓

b. $a = 1, b = 4, c = -5$

c. $a = 2, b = -8, c = 7$

d. $a = -1, b = 4, c = -5$

The vertex of the parabola $y = ax^2 + bx + c$ is at $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$. We are given that the vertex is at $(2, -1)$. Therefore, $-\frac{b}{2a} = 2$, which implies $b = -4a$. The vertex form of the quadratic equation is $y = a(x - h)^2 + k$, where (h, k) is the vertex. So, $y = a(x - 2)^2 - 1$. The graph passes through $(4, 3)$. Substituting this point into the equation: $3 = a(4 - 2)^2 - 1 \Rightarrow 3 = a(2)^2 - 1 \Rightarrow 3 = 4a - 1 \Rightarrow 4a = 4 \Rightarrow a = 1$. Si.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

a. 1

b. **2 ✓**

c. 3

d. 4

We are given the function $f(x) = a \cdot b^x$ and two points it passes through: $(1, 12)$ and $(3, 108)$. Using the first point $(1, 12)$, we have: $12 = a \cdot b^1$ (Equation 1) Using the second point $(3, 108)$, we have: $108 = a \cdot b^3$ (Equation 2) To find a and b, we can divide Equation 2 by Equation 1: $\frac{108}{12} = \frac{a \cdot b^3}{a \cdot b^1}$ $9 = b^2$ Taking the square root of both sides, and assuming $b > 0$ for exponential functions: $b = 3$ Now, substitute the value of $b = 3$ back into E.

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